Chapter 8 exercises

Vectors are columns unless stated otherwise. For each coordinate matrix, state the domain basis, codomain basis, and coordinate order.

CH08-E1

Decide linearity for each map. Prove it or give a counterexample.

1. $T_1(x, y) = (2x - y, x + 3y)$.
2. $T_2(x, y) = (x + y, 1)$.
3. $T_3 : \mathcal{P}_2 \to \mathcal{P}_2, T_3(p)(t) = tp'(t)$.
4. $T_4(f) = f^2$ for $f \in C^\infty(\mathbb{R})$, with pointwise multiplication.

CH08-E2

Define $T : \mathbb{R}^3 \to \mathbb{R}^2$ by

$$T(x, y, z) = (x + 2y - z,\ x - y + 2z).$$

1. Find its standard matrix.
2. Find a kernel basis and nullity.
3. Find an image basis and rank.
4. Verify rank-nullity.
5. Decide injectivity and surjectivity.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

CH08-E3

Let $T : \mathbb{R}^2 \to \mathbb{R}^3$ have standard matrix

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Input basis:

$$B_{\text{in}} = \{(1, 1)^T, (1, -1)^T\}.$$

Output basis:

$$B_{\text{out}} = \{e_1, e_2, e_3\}.$$

1. Find its matrix in the specified input basis and standard output basis.
2. If the output basis changes to

$$(1, 0, 0)^T, (1, 1, 0)^T, (1, 1, 1)^T,$$

find the new matrix. 3. Verify the coordinate conversion formula.

Chapter 8 exercises (continued)

CH08-E4

Let

$$A = \begin{bmatrix} 4 & -2 \\ 1 & 1 \end{bmatrix}.$$

1. Find eigenvalues and eigenvectors.
2. Construct B making $B^{-1}AB$ diagonal.
3. Compute the diagonal matrix and check trace and determinant.
4. Describe the effect of multiplying each eigenvector by a nonzero scalar.

CH08-E5

Consider derivative operator $D\colon \mathcal{P}_2 \to \mathcal{P}_1$ with monomial bases.

1. Give its coordinate matrix.
2. Find kernel and image and verify rank-nullity.
3. Change the domain basis to $1, t-1, (t-1)^2$ and find the new matrix.
4. Explain why these rectangular matrices are not similar.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

CH08-E6

Give

$$N = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

1. Find the eigenvalue and eigenvectors of N.
2. Find a generalized eigenvector w whose image Nw is an ordinary nonzero eigenvector.
3. Give a Jordan form and basis matrix satisfying $N = BJB^{-1}$.
4. Explain why N is not diagonalizable despite having only one distinct eigenvalue.

CH08-E7

Fix the cyclic permutation direction as

$$P = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}, \qquad P(x_0, x_1, x_2, x_3)^T = (x_1, x_2, x_3, x_0)^T.$$

and consider the Fourier basis vectors

Chapter 8 exercises (continued)

$$v_\lambda = (1, \lambda, \lambda^2, \lambda^3)^T, \qquad \lambda \in \{1, i, -1, -i\}.$$

1. Prove $Pv_\lambda = \lambda v_\lambda$.
2. Prove independence of these four vectors.
3. For $C = c_0 I + c_1 P + c_2 P^2 + c_3 P^3$, give the diagonal entries in this Fourier basis.
4. Explain why a real P can have complex eigenvalues.

CH08-E8

Let

$$B_\epsilon = \begin{bmatrix} 1 & 1 \\ 0 & \epsilon \end{bmatrix}.$$

1. Prove the columns of B_ϵ form a basis for every $\epsilon > 0$.
2. Find $\kappa_2(B_\epsilon)$ and its limit as $\epsilon \to 0^+$.
3. Describe the angle between the basis vectors.
4. Bound reconstruction error when coordinate error is 10^{-8}. Distinguish a norm bound from a bound on each coordinate, and interpret the conditioning.

CH08-E9

Let $T(x, y) = (x + y, x - y, 2x)$, $S(x, y, z) = (x + z, y - 2z)$.

1. Find their standard matrices.
2. Find the standard matrix of $S \circ T$ and explain the multiplication order.
3. For $(x, y) = (1, 2)$, compare direct substitution with the composition matrix.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

References and attribution

Gilbert Strang. Introduction to Linear Algebra, 5th Edition, 2016.

Original educational synthesis by Lin Cing. Original material is licensed under CC BY 4.0. The referenced textbook is not distributed with these notes.

https://github.com/xinhi6716/computer-science-self-study